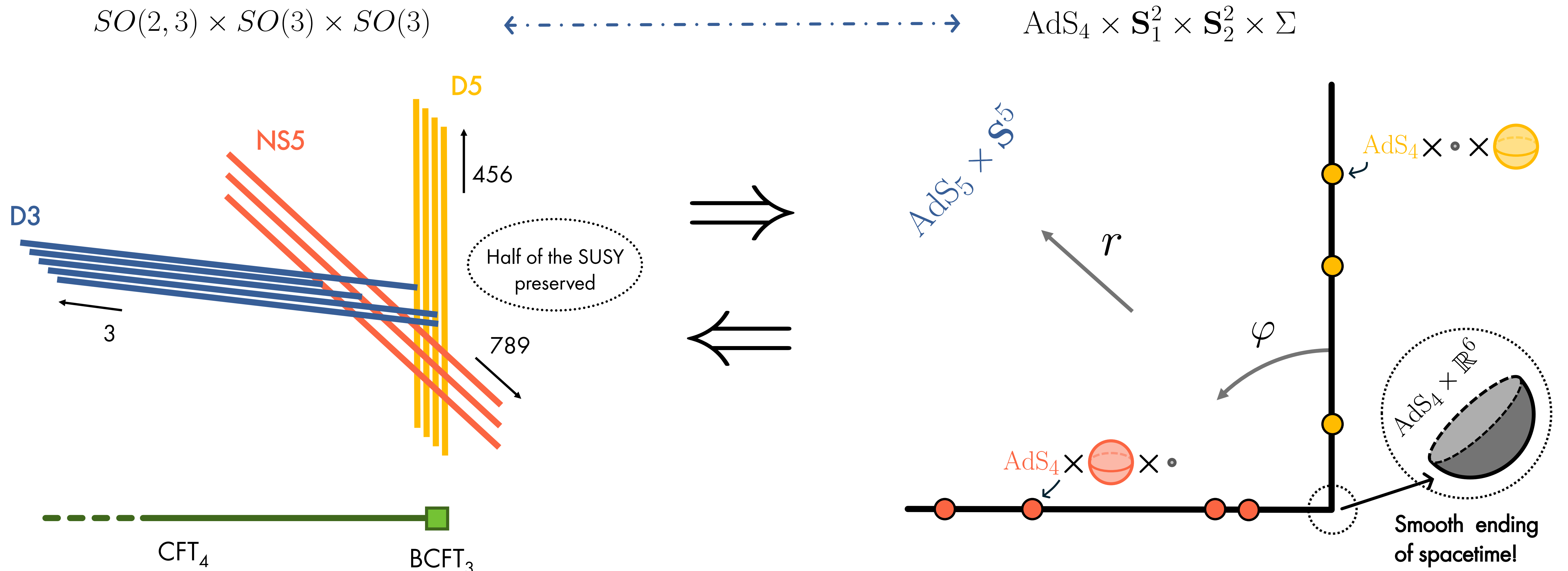


End of the World branes in $AdS_5 \times S^5$?

According to the Cobordism Conjecture [1], every consistent Quantum Gravity theory must admit a cobordism to nothing...
...or in other words, it must admit **End of the World branes**

We present the ETW branes for $AdS_5 \times S^5$, using the solutions of [2,3], and find:

- Karch-Randall branes in String Theory
- A new double scaling limit which isolate the ETW brane
- The 5-brane origin of the degrees of freedom of the ETW



KARCH-RANDALL BRANE IN STRING THEORY

The 5-branes break the $SO(6)$ invariance of $AdS_5 \times S^5$ to $SO(3) \times SO(3)$. We need to find a way to simplify the solution to compare with standard ETW branes. We start with the full 10D SUGRA metric:

$$ds^2 = f_4^2 ds_{AdS_4}^2 + f_1^2 ds_{S_1^2}^2 + f_2^2 ds_{S_2^2}^2 + 4\rho^2 (dr^2 + r^2 d\varphi^2)$$

and parametrize the volume of the 2-spheres introducing radions, so we have

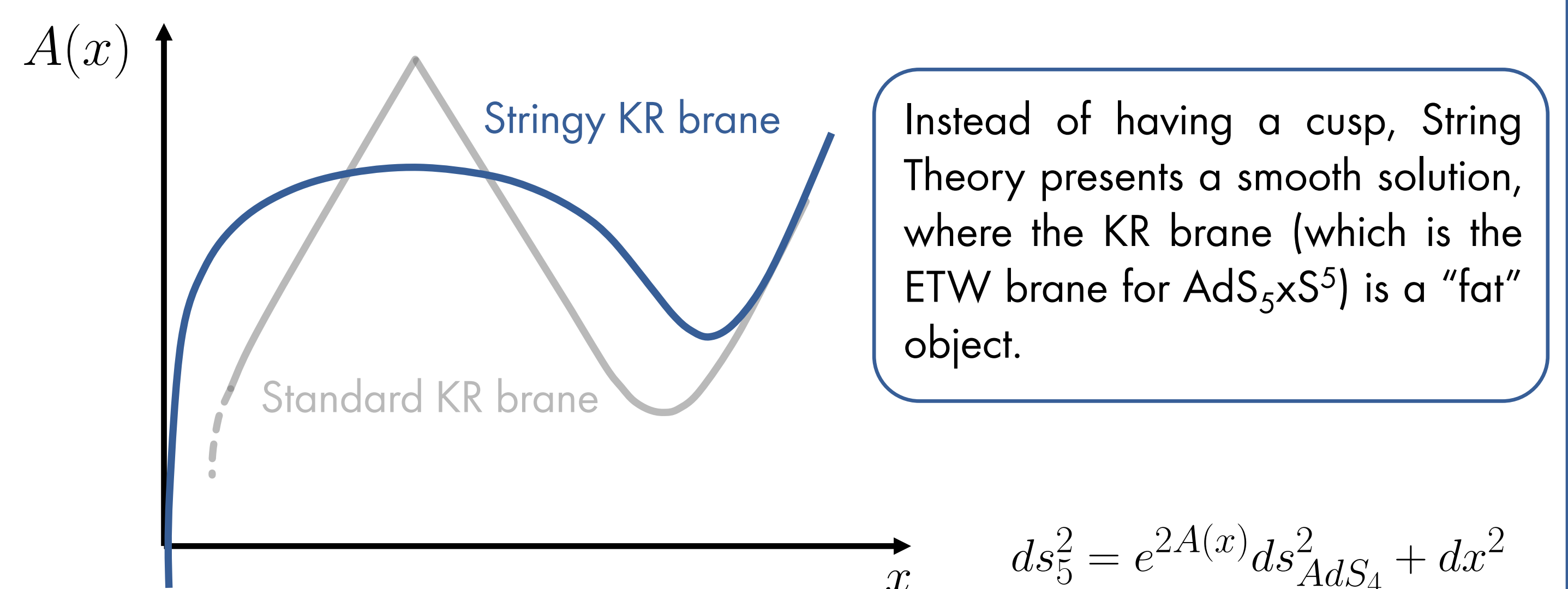
$$ds_6^2 = \sqrt{f_1^2 f_2^2} [f_4^2 ds_{AdS_4}^2 + 4\rho^2 (dr^2 + r^2 d\varphi^2)]$$

We then truncate onto the zero mode of the KK tower (the $SO(6)$ invariant sector), which parametrizes the volume of the S^5

$$g_{\mu\nu}(\varphi) = \sum_i f_i(\varphi) \hat{g}_{\mu\nu}^i \quad f_0(\varphi) = -\frac{2}{\sqrt{\pi}} \sin 2\varphi \quad \hat{g}_{\mu\nu} = \int_{\pi/2}^{\pi} f_0(\varphi) g_{\mu\nu}(r, \varphi) d\varphi$$

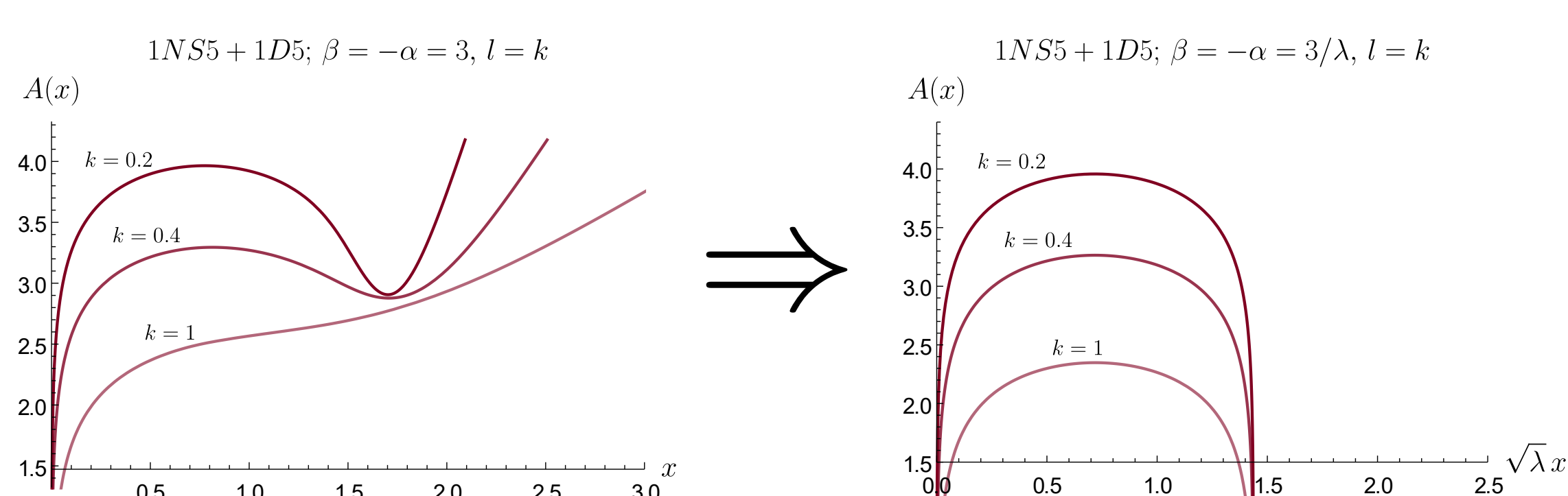
Finally, we arrive to the desired AdS_4 foliation of AdS_5 , which allows us to make contact with the literature in ETW in AdS , and the Karch-Randall solutions, which are branes that **localize gravity** in their worldvolumes [4]:

$$ds_5^2 = \hat{f}_4^2 ds_{AdS_4}^2 + 4\hat{\rho}^2 dr^2$$



DOUBLE SCALING LIMIT

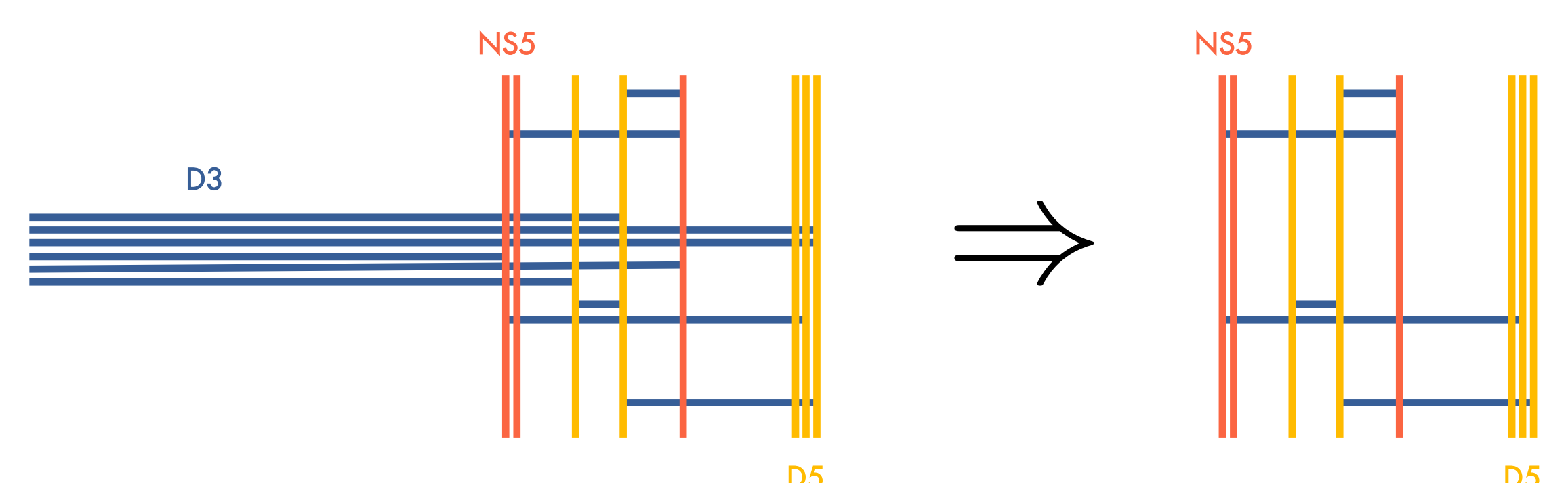
We can take a double scaling limit of the solution which "zooms" on the ETW. This is equivalent to decouple the AdS_5 throat



This makes the geometry compact, which, in fact, provides the gravity dual of the 3d BCFT (this is related to **wedge holography**, ask for more info!).

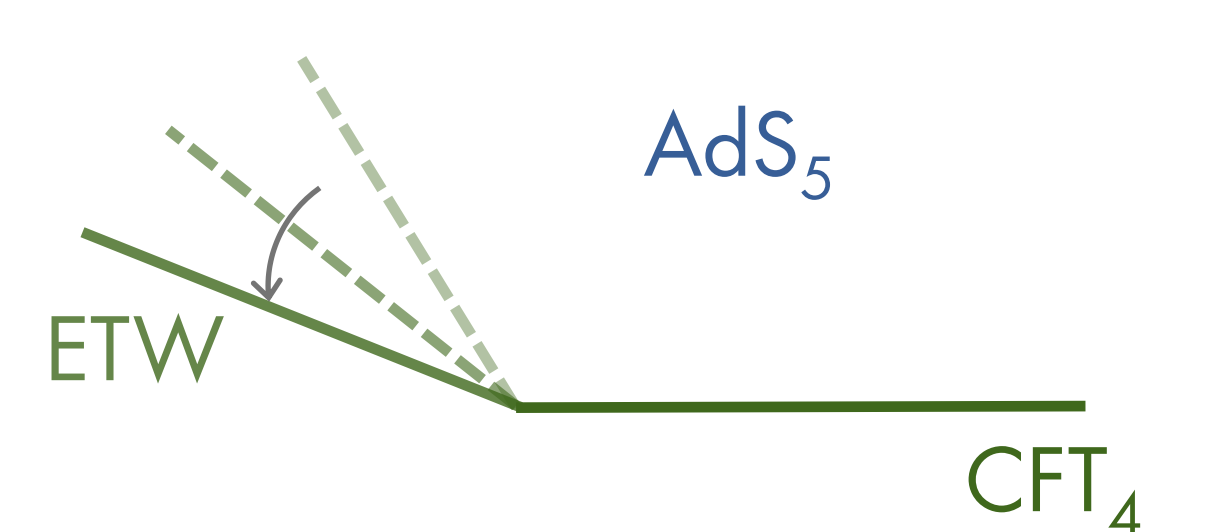
We realized that, to take the double scaling limit (and to have localize gravity), we need both D5- and NS5-branes.

5-BRANE DOF OF THE ETW BRANE



The need of both D5- and NS5-branes is a hint that we need the D3-branes suspended between them -that are "frozen" according to Hanany-Witten- to localize gravity in the ETW brane.

The more 5-branes we add, the more the ETW approaches the holographic boundary in Poincaré coordinates, and the smaller the mass of the graviton. The ETW **"reconstructs"** the AdS_5 missing.



This poster is based in [5] and work in progress with Ángel Uranga

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References:

- [1] J. McNamara and C. Vafa, *Cobordism Classes and the Swampland*, [1909.10355].
- [2] E. D'Hoker, J. Estes, and M. Gutperle, *Exact half-BPS Type IIB interface solutions*, [0705.0024].
- [3] O. Aharony, L. Berdichevsky, M. Berkooz, and I. Shamir, *Near-horizon solutions for D3-branes ending on 5-branes*, [1106.1870].
- [4] A. Karch and L. Randall, *Locally localized gravity*, [hep-th/0011156].
- [5] J. Huertas and A. M. Uranga, *Aspects of dynamical cobordism in AdS/CFT*, [2306.07335].